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Bifurcation and Resonance in a Model for Bursting Nerve Cells

Richard E. Plant

Department of Mathematics, University of California, Davis, CA 95616, USA

Abstract. In this paper we consider a model for the phenomenon of bursting in nerve cells. Experimental evidence indicates that this phenomenon is due to the interaction of multiple conductances with very different kinetics, and the model incorporates this evidence. As a parameter is varied the model undergoes a transition between two oscillatory waveforms; a corresponding transition is observed experimentally. After establishing the periodicity of the subcritical oscillatory solution, the nature of the transition is studied. It is found to be a resonance bifurcation, with the solution branching at the critical point to another periodic solution of the same period. Using this result a comparison is made between the model and experimental observations. The model is found to predict and allow an interpretation of these observations.

Key words: Bifurcation – Bursting

I. Introduction

Many nerve cells in both vertebrate and invertebrate nervous systems display a behavior known as *bursting*. This consists of the generation of membrane action potentials in regular sequences of "bursts" which are separated by periods of inactivity during which the cell membrane may be hyperpolarized. The reader who is unfamiliar with bursting may refer to Fig. 1, which is the numerical solution of a model developed to simulate this phenomenon. Among the more commonly studied nerve cells which display this behavior are abdominal ganglion cells of molluscs [5, 12, 22], cardiac pacemaker and stomatogastic ganglion cells of crustacea [21, 26], and penicillin treated pyramidal cells of the vertebrate hippocampus [24]. Bursting is important experimentally for two reasons. First, it is a relatively simple neural pattern and yet one which is involved in the control of many physiological and behavioral activities such as blood circulation, respiration, mastication, copulation and locomotion. Second, nerve cells involved in epileptic seizure activity have been observed to burst during interictal (i.e. between seizures) states. A knowledge of the mechanisms underlying bursting may therefore lead to an increased understanding of the causes of convulsive behavior.

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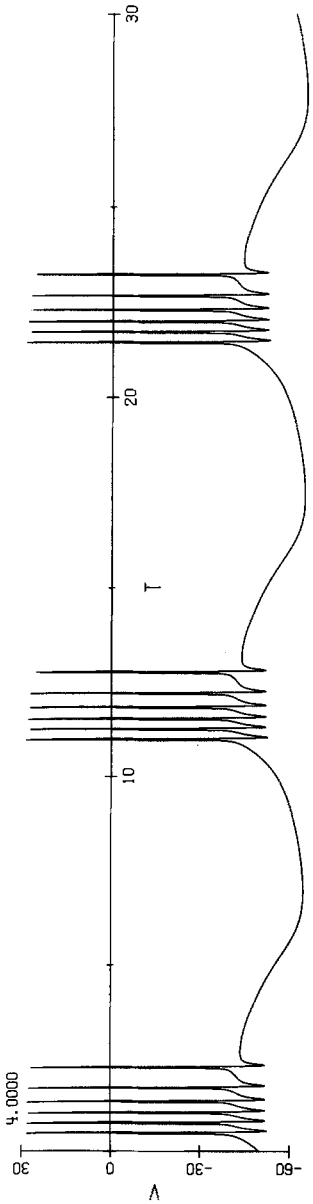


Fig. 1

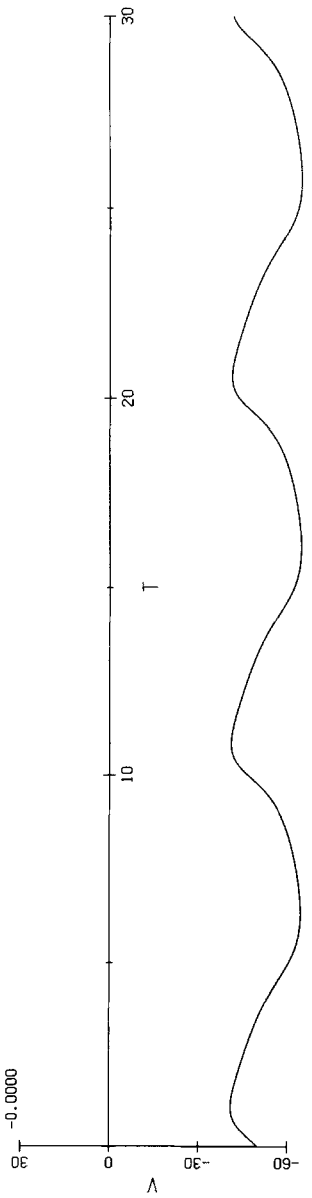


Fig. 2

It has been shown [3] that if the speed of propagated action potentials is included as a parameter, then models involving two interacting variable conductances of the type described by Hodgkin and Huxley [8] may display bursting. However, there is strong evidence that in each of the experimental preparations listed above the physical properties of the system are more complex [5, 12, 14, 15, 23]. Specifically, the bursting waveform evidently results from the interaction of the spike generating "Hodgkin-Huxley" conductances with another pair of conductances which have considerably slower kinetics. In certain preparations such as the *Aplysia* R15 cell the spike generating sodium conductance (i.e. g_{Na} in the Hodgkin-Huxley equations) may be abolished by application of the nerve poison tetrodotoxin (TTX). When this is done the action potentials disappear and an underlying "slow wave" is unmasked [14, 22]. The reader may refer to Fig. 2, which is a simulation of this situation, to gain insight into the slow wave. It is evident that in the unpoisoned preparation the action potentials "ride" this slow wave (cf. [14, Fig. 3]).

There is strong evidence that, at least in the preparations described above, a substantial fraction of the inward current of the slow conductance is carried by calcium ions [5, 23]. In addition to carrying inward current, calcium ions are involved in the activation of the slow outward conductance [15, 23], whose current is carried by potassium ions. In terms of a Hodgkin-Huxley type formalism, the voltage component of a model for such a system may be written

$$\dot{V} = [g_I(t) + g_T(t)][V_I - V] + [g_K(t) + g_P(t)][V_K - V] + g_L[V_L - V] \quad (1.1)$$

where the dot denotes differentiation with respect to t; V is the membrane voltage; g_I, g_T, g_K, g_P, and g_L are respectively the fast inward, slow inward, fast outward, slow outward, and leakage conductances; and V_I, V_K, and V_L respectively the mixed sodium-calcium, potassium, and leakage equilibrium potentials. It should be noted that equation (1.1) ignores several phenomena such as other conductances and steady active transport currents which are also observed in these preparations. Equation (1.1) assumes unit capacitance so that the dimensions of the conductance functions $g_i(t)$ are mmho/μf. This convention will be assumed throughout the sequel.

Mathematical models based on concepts similar to equations (1.1) have been developed [16, 23, 24]. Indeed, the existence and many of the properties of the slow conductances had been predicted as a consequence of a modelling study before these conductances had been experimentally measured [17].

In terms of the formalism of equation (1.1), the difference between the models of Figs. 1 and 2 is that in Fig. 2 $g_I(t)$ is zero, while in Fig. 1 it is not. It appears that the transition from the model of Fig. 2 to Fig. 1 represents the bifurcation from one periodic solution to another, with the bifurcation parameter being contained in the function $g_I(t)$. This bifurcation is of interest for two reasons. First, a knowledge of the roles of the various components of the model in the bifurcation may be useful in determining the corresponding role of these components in the transition to convulsive behavior. Second, the phenomenon of bifurcation of periodic solutions is of interest in its own right and the bursting model represents a relatively simple example of this form of bifurcation.

The present article is devoted to a study of the periodic solutions represented in Figs. 1 and 2. The equations which are considered are based on a simplified version of the system studied in [16]. These equations are

$$\begin{aligned}
\dot{V} &= [g_I x_I^3 y_I + g_T x_T][V_I - V] + \left[g_K x_K^4 + \frac{g_P c}{K_P + c}\right][V_K - V] + g_L[V_L - V],\\
\dot{x}_i &= [s_i(V) - x_i]/\tau_{xi}, \qquad i = I, T, K,\\
\dot{y}_I &= [z_I(V) - y_I]/\tau_{yI},\\
\dot{c} &= \rho[K_c x_T(V)[V_{\mathrm{Ca}} - V] - c].
\end{aligned} \tag{1.2}$$

The functions $s_i(V)$ are s-shaped, that is, monotonically increasing continuously differentiable functions with exactly one inflection point and satisfying $0 < s_i(V) < 1$, $\lim_{V \to -\infty} s_i(V) = 0$, $\lim_{V \to \infty} s_i(V) = 1$. The function $z_I(V)$ is z-shaped; it satisfies $z_I(V) = 1 - s(V)$ where $s(V)$ is an s-shaped function as defined above. The variable c represents the calcium ion concentration at the inner face of the membrane. It is increased by inward calcium current at a rate $\rho K_c x_T[V_{\mathrm{Ca}} - V]$ where V_{Ca} is the equilibrium potential for calcium (assumed constant), and it is decreased at a rate ρc. The variable c enters the $\dot{V}$ equation in the form $c/(K_P + c)$; this represents a first order exchange process in quasi steady state with a dissociation constant K_P. The relaxation times τ_{xi} and τ_{yI}, which may depend on V, satisfy

$$\tau_{xI} \ll \tau_{yI},\ \tau_{xT},\ \tau_{xK} \ll \rho^{-1}. \tag{1.3}$$

A full discussion of the biological evidence for this model is given in references [16] and [18]. Broadly similar models have been developed independently by Thompson [23], and Traub and Llinas [24].

The actual system considered in the study is modified from equation (1.2). First, noting inequality (1.3), we make the standard simplification of replacing the variable x_I in the first of equations (1.2) with its steady state value $s_I(V)$. Second, again noting inequality (1.3), we multiply the left-hand sides of the equations for $\dot{x}_T$, $\dot{x}_K$ and $\dot{y}_I$ by a parameter ζ, which takes on the value $\zeta = 1$ in equations (1.2). This results in the following system:

$$\begin{aligned}
\dot{V} &= [g_I s_I^3(V) y_I + g_T x_T][V_I - V] + [g_K x_K^4 + g_P c(K_P + c)^{-1}][V_K - V]\\
&\quad + g_L[V_L - V],\\
\zeta \dot{x}_i &= [s_i(V) - x_i]/\tau_{xi}, \qquad i = T, K,\\
\zeta \dot{y}_I &= [z_I(V) - y_I]/\tau_{yI},\\
\dot{c} &= \rho[K_c x_T(V_{\mathrm{Ca}} - V) - c].
\end{aligned} \tag{1.4}$$

This is the system studied in the present paper.

Figure 1 shows the V component of the numerical solution of equations (1.4) with $\zeta = 1$ and the parameters and functions having the values given in the Appendix. The number in the upper left corner of Fig. 1 is the value of g_I. This figure represents the bursting waveform of the system under normal conditions. Qualitatively the waveform is quite similar to that observed, for example, in the

crayfish stomatogastric ganglion [21], the mantis shrimp cardiac ganglion [26], and the L6 cell of the *Aplysia* abdominal ganglion. It does not have the particular "parabolic" shape of the *Aplysia* R15 cell (cf. [18, p. 241]). Figure 2 shows the V component of the numerical solution for the same system, but with $g_I = 0$. This is the model for the nerve cell bathed in TTX (cf. [14, 22].

The remainder of the paper is organized as follows. In Section 2 we establish the existence and properties of the periodic solution demonstrated numerically in Fig. 2 for the system with $g_I = 0$. The periodic solution is established for sufficiently small values of the parameter ζ by demonstrating the existence of a Hopf bifurcation, with bifurcation parameter g_T, for a second order system obtained by setting $\zeta = 0$. The periodic solution of the full system is then established by invoking a theorem of Flatto and Levinson [4]. The results of this section are similar to earlier results of Carpenter [2], Hsu and Kazarinoff [9], and Troy [25], but the point of view is different. In Section 3 the bifurcation from one form of periodic solution to another is studied numerically. The bifurcation parameter used in this section is the parameter g_I. Numerical results indicate that the bifurcation is a resonance case, with a single real eigenvalue of the Jacobian of the Poincaré map crossing the unit circle.

Bifurcation theoretic studies may be useful since they represent transitions from one form of solution to another, and the corresponding transitions in the real system may often be observed experimentally. Since the properties of the transition may be predicted based on the nature of the bifurcation, these studies provide a basis for comparison of model and experiment. In Section 4, such a comparison is presented.

2. Periodic Solutions with $g_I = 0$

In order to study bifurcations from a periodic solution, it is necessary to establish the existence and stability of that periodic solution. That is the object of this section. We study a reduced version of equations (1.4) obtained by assigning the value zero to g_I. We begin by setting ζ equal to zero. The resulting singularly perturbed system is

$$\dot{V} = g_T s_T(V)[V_I - V] + [g_K s_K^4(V) + g_P c(K_P + c)^{-1}][V_K - V] + g_L[V_L - V],$$
$$\dot{c} = \rho[K_c s_T(V)(V_{Ca} - V) - c]. \tag{2.1}$$

Let us define functions G and H by setting the left-hand sides of equations (2.1) equal to zero. Letting

$$F(V; g_T) = [V - V_K]^{-1}[g_T s_T(V)(V_I - V) + g_L(V_L - V)] - g_K s_K^4(V) \tag{2.2}$$

we have for the equation $\dot{V} = 0$,

$$c = G(V; g_T) = K_P F(V; g_T)/(g_P - F(V; g_T)) \tag{2.3}$$

and for the equation $\dot{c} = 0$,

$$c = H(V) = K_c s_T(V)(V_{Ca} - V). \tag{2.4}$$

The equilibrium points of system (2.1) are given by the values of V such that $H(V) = G(V; g_T)$. There is an unstable equilibrium point defined by an intersection

for $V_I < V < V_{Ca}$. Provided V is less than V_I initially, the solution $V(t)$ must remain less than V_I for all t so that this equilibrium point is of no consequence. Indeed, the existence of this equilibrium point is due to the simplifying assumption that V_{Ca} is constant (actually V_{Ca} should depend on c).

Consider now values of V in the range $V_K \leqslant V \leqslant V_I$. We wish to assign certain properties to the functions G and H. Differentiating equation (2.3) with respect to V yields

$$G'(V; g_T) = g_P K_P F'(V; g_T)/(g_P - F(V; g_T))^2 \tag{2.5}$$

where the prime denotes differentiation with respect to V. Thus for $F \neq g_P$ the sign of G' is the same as that of F'. Differentiating equation (2.2) yields

$$F'(V; g_T) = g_T s_T'(V)\frac{[V - V_I]}{[V - V_K]} - s_T(V)\frac{[V_I - V_K]}{[V - V_K]^2} - g_L\frac{[V_L - V_K]}{[V - V_K]^2} - [s_K^4(V)]'. \tag{2.6}$$

It follows from our assumptions on the g_i, the V_i, and the functions $s_i(V)$ that all the quantities on the right-hand side are positive for all $V \in (V_K, V_I)$. Moreover, the s-shape of the function $s_T(V)$ implies that $s_T'(V)$ will be approximately zero except where $s_T(V)$ moves from zero to one. Thus we expect $F'(V; g_T)$ to be negative as V approaches V_K or V_I, but possibly to be positive for intermediate values of V. We make the following hypotheses:

(H_1) Let $F(V^*) = g_P$. Then there exist two values of V, denoted by V_a and V_b, with $V^* < V_a < V_b < V_I$, such that for some $g_T, F'(V; g_T) > 0$ for $V_a < V < V_b$. Moreover, the inequality $V_K < V_a$ is satisfied.

(H_2) There is a unique $V \in (V_K, V_I)$, denoted by V_c, with $V_a < V_c < V_b$, such that $G(V_c) = H(V_c)$.

Figure 3 shows the functions $G(V, g_T)$ and $H(V)$, for several values of g_T, with values of the parameters given in the Appendix. It is evident that for some g_T between 0.05 and 0.15, and in particular for $g_T = 0.01$, hypotheses H_1 and H_2 are satisfied. We will now show that the assumptions of Section 1 on the parameters, together with hypotheses H_1 and H_2, are sufficient to guarantee the existence of a periodic solution provided ρ is sufficiently small. This will be accomplished by means of the Hopf bifurcation theorem.

If we let $\eta_1 = V - V_c, \eta_2 = c - c_c$, where (V_c, c_c) is the unique equilibrium point of equations (2.1) such that $V_a < V_c < V_b$, then we may rewrite equations (2.1) as

$$\dot{\eta}_1 = f_1(\eta_1, \eta_2, g_T), \qquad \dot{\eta}_2 = f_2(\eta_1, \eta_2, g_T) \tag{2.6}$$

where f_1 and f_2 are defined in the obvious way. Note that V_c and c_c depend on g_T. Let $A(g_T)$ be the Jacobian matrix $\partial(f_1, f_2)/\partial(\eta_1, \eta_2)$ evaluated at $(\eta_1(g_T), \eta_2(g_T)) = (0, 0)$, and let $\lambda(g_T) = \alpha(g_T) \pm i\omega(g_T)$ be the eigenvalues of A. Then the statement by Alexander and Yorke [1] of the Hopf bifurcation theorem implies that equations (2.6) will have a Hopf bifurcation at $g_T = g_T^*$ provided (i) $\alpha(g_T^*) = 0$ and $\omega(g_T^*) \neq 0$, (ii) $\alpha(g_T)$ changes sign as g_T passes through g_T^*.

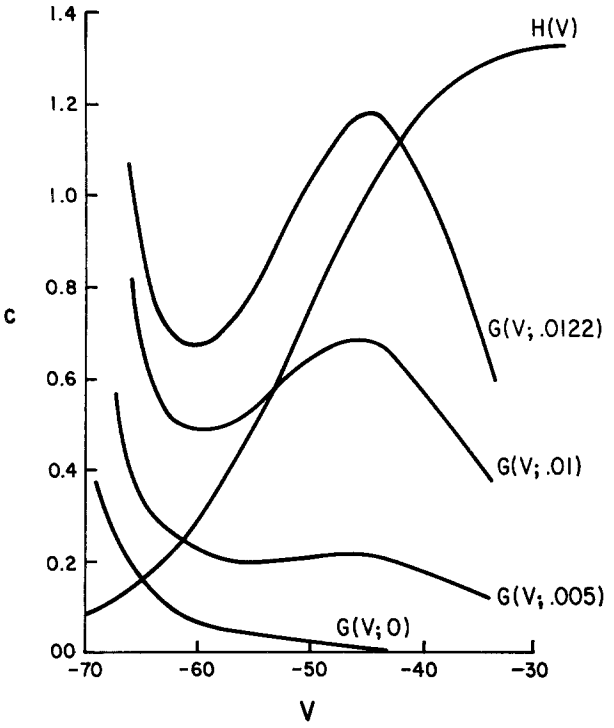


Fig. 3

It follows from the definitions of f_1, f_2, G, and H that A is given by

$$A = \begin{bmatrix} -\dfrac{\partial f_1}{\partial \eta_2} G' & \dfrac{\partial f_1}{\partial \eta_2} \\ \rho H' & -\rho \end{bmatrix} \tag{2.7}$$

where all quantities are evaluated at $\eta_1 = \eta_2 = 0$. The trace and determinant of A are given respectively by

$$\sigma(g_T) = -\frac{\partial f_1}{\partial \eta_2}(0,0)G'(V_c) - \rho, \tag{2.8a}$$

$$\Delta(g_T) = \rho \frac{\partial f_1}{\partial \eta_2}(0,0)[G'(V_c) - H'(V_c)]. \tag{2.8b}$$

It follows from equations (2.1) and (2.6) that

$$\frac{\partial f}{\partial \eta_2}(0,0) = \frac{g_P K_P}{(K_P + c_c)^2}(V_K - V_c), \tag{2.9}$$

so that by hypotheses H_2 we have $(\partial f/\partial \eta_2)(0,0) < 0$. Furthermore, by equations (2.5) and (2.6) we have $G'(V;0) < 0$, and therefore $\sigma(0) < 0$ for all $V \in (V^*, V_a)$. By hypotheses H_1 and H_2 there exists a $\tilde{g}_T$ such that $G'(V_c; \tilde{g}_T) > 0$, and therefore

$-(\partial f_1/\partial \eta_2)(0,0)G'(V_c;\tilde{g}_T) > 0$. Therefore by equation (2.8a) and the continuity of $(\partial f_1/\partial \eta_2)(0,0)$ and G' in g_T there is a g_T^* in $(0,\tilde{g}_T)$ such that for ρ a sufficiently small constant, $\sigma(g_T^*) = 0$, and for some $g_T > g_T^*$, $\sigma(g_T) > 0$.

From equation (2.3) and hypothesis H_1 it follows that

$$\lim_{\substack{V \to V^* \\ V > V^*}} G(V; g_T) = \infty \tag{2.10}$$

for all $g_T \in [0,\tilde{g}_T]$. Therefore by hypothesis H_2 we must have $G'(V_c;g_T) < H'(V_c;g_T)$ since otherwise there would have to be at least two intersections of G and H. Therefore by equation (2.8b) we have $\Delta(g_T) > 0$ for $g_T \in [0,\tilde{g}_T]$. In summary, there must exist a g_T^* in $(0,\tilde{g}_T)$ such that for sufficiently small ρ the following are satisfied:

i) $$\alpha(g_T^*) = \tfrac{1}{2}\sigma(g_T^*) = 0.$$

ii) $\alpha(g_T)$ changes sign from negative to positive as g_T increases through g_T^*.

iii) $$\omega_0 = \sqrt{\Delta(g_T^*)} = \rho\left(\frac{H'(V_c;g_T^*)}{G'(V,g_T^*)} - 1\right)^{1/2} > 0.$$

Therefore by Theorem A of Alexander and Yorke [1] there must exist a bifurcation at g_T^* and we have proved the following.

Theorem 1. *Let the constants, parameters and functions satisfy conditions* (A.1) *of the appendix with* $g_I = 0$ *and be such that hypotheses* (H_1) *and* (H_2) *are satisfied. Then there exists a* $g_T^* > 0$ *such that for sufficiently small positive values of* ρ, *at* $g_T = g_T^*$ *the system* (2.1) *has a Hopf bifurcation to a periodic solution.*

We must now consider the stability of the bifurcation. Hsu and Kazarinoff [9] have given a formula for the computation of the stability of the bifurcating solution. Since this formula involves second and third derivatives of the functions in equations (2.1) we cannot obtain analytic conditions for the stability or instability of the bifurcation. We therefore compute the stability of the bifurcation for the special case of values given in the Appendix. Carrying out the computations of $\alpha(g_T)$ and $\omega(g_T)$ based on equations (2.7) and (2.8) yields zero values for $\alpha(g_T)$ at $g_T = 0.007856$ and $g_T = 0.014559$. At these values we have $\alpha'(0.007856) = 1.22$ and $\alpha'(0.011456) = 0.511$, computed numerically to three figures. Also $\omega_0(0.007856) = 1.99 \times 10^{-3}$ and $\omega_0(0.011456) = 1.66 \times 10^{-3}$. Therefore we have Hopf bifurcations at these two values of g_T. We may evaluate the functions in Theorem 2.3 of Hsu and Kazarinoff and thereby obtain the values for μ_2

$$\mu_2(0.007856) = 3.71 \times 10^{-3} \qquad \text{and} \qquad \mu_2(0.011456) = 4.96 \times 10^{-2}$$

and therefore the bifurcation at $g_T = 0.007856$ is supercritical, while the bifurcation at $g_T = 0.011456$ is subcritical.

Figure 4 shows a plot of the amplitude (i.e. maximum minus minimum) vs. g_T of the η_1 component of equations (2.6). This was obtained numerically using the GEAR subroutine package as described in Section 3. It follows from Theorem A of Alexander and Yorke that the bifurcating periodic solution must be continued as g_T increases from 0.007856 until one of four events takes place: either the periodic

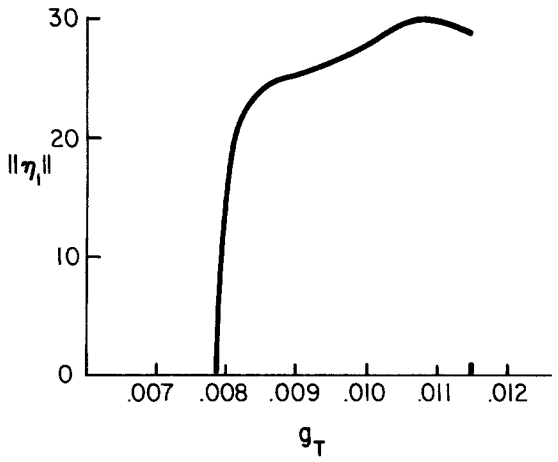


Fig. 4

solution undergoes another bifurcation, the amplitude of the periodic solution tends to infinity, the period of the solution tends to infinity, or the bifurcation parameter tends to infinity. Numerical evidence indicates that neither the amplitude nor the period grow to infinity over the range of parameter values shown in Fig. 3, therefore we have strong evidence for the existence of a periodic solution over this range of parameter values. It is worth noting that the existence of this periodic solution could also be obtained by an application of the Poincaré-Bendixson theorem.

The orbital stability of this periodic solution may be determined numerically by computing the single nonzero characteristic exponent. Such computations indicate that the sign of this exponent is negative along the solid curve in Fig. 3, and therefore the family of periodic solutions represented by this curve is stable. There is presumably a family of unstable periodic solutions which pass from the bifurcation point at $g_T = 0.011456$ to intersect the stable solutions at some value $g_T > 0.011456$. However, these unstable solutions must increase in amplitude very quickly since no stable periodic numerical solutions of equations (2.1) were found for $g_T = 0.012$.

Having demonstrated the existence of a periodic solution for the reduced system (2.1), we now consider the full system

$$\begin{aligned} \dot{V} &= g_T x_T[V_I - V] + [g_K x_K^4 + g_P c(K_P + C)^{-1}][V_K - V] + g_L[V_L - V], \\ \dot{c} &= \rho[K_c x_T(V_{\mathrm{Ca}} - V) - c], \\ \zeta \dot{x}_i &= [s_i(V) - x_i]/\tau_{xi}, \qquad i = T, K. \end{aligned} \tag{2.11}$$

For $\zeta = 0$ we have $x_i = s_i(V)$, $i = T, K$, and this gives us equations (2.1). We have established that the periodic solution of equation (2.1) has negative real part (the other characteristic exponent having zero real part). Moreover, if we let the right-hand sides of the equations in $\dot{x}_T$ and $\dot{x}_K$ in equation (2.11) define a function $g(V, c, x_T, x_K)$ then we have

$$\frac{\partial g}{\partial(x_T, x_K)} = \begin{bmatrix} -\tau_{xT}^{-1} & 0 \\ 0 & -\tau_{xK}^{-1} \end{bmatrix}.$$

Since the τ_{xi} are strictly positive, we may apply the theorem of Flatto and Levinson [4] to obtain a periodic solution to equation (2.11) for sufficiently small ζ. Of course we have not *proved* the existence of such a solution since our results are computational; in addition there is no guarantee that the value $\zeta = 1$ is sufficiently small to ensure a periodic solution. We shall not carry this matter further except to note that numerical integration of equations (2.11) produces a periodic solution to the accuracy of the integration scheme.

3. Computation of the Bifurcation of the Periodic Solution

Having established, at least numerically, the existence of a periodic solution to the system (1.4) with $\zeta = 1$ and $g_I = 0$, we now use g_I as a bifurcation parameter to study the transition from solutions of the form of Fig. 2 to those of the form of Fig. 1. We shall fix the value of g_T at $g_T = 0.01$ throughout this section and only allow the value of g_I to vary. In order to study the bifurcation from a periodic orbit we must examine the behavior of the eigenvalues of the Jacobian of the Poincaré map.

Let $p(t)$ be a periodic solution to equations (1.4). The orbit Γ of $p(t)$ is a closed curve in $\mathbb{R}^5$. Let Σ be a smooth 4-dimensional surface in the neighborhood of a point on Γ transversal to Γ. Then the Poincaré map of (1.4) is the map $\Psi: \Sigma \to \Sigma$ which maps a point q in Σ into the first point where the solution to (1.4) intersects Σ again. If the intersection of Γ with Σ is the point q then $q = q_0$ is a fixed point of Ψ. The eigenvalues of the Jacobian of $\Psi(q_0)$ are the Floquet multipliers of the period solution $p(t)$ [13] so that if $p(t)$ is orbitally stable then these eigenvalues have modulus less than 1. As one or more eigenvalues pass through the unit circle we expect a breakdown of stability, or bifurcation, to occur.

The nature of this bifurcation has been studied by several authors, including Iooss and Joseph [11], Ruelle and Takens [19], and Sacker [20]. Let $\lambda(\mu) = r(\mu)e^{i\theta(\mu)}$ be an eigenvalue such that $r(0) = 1$ and assume that all other eigenvalues except the complex conjugate of λ have modulus less than unity. Assume that $r'(0) \neq 0$. Then a bifurcation occurs at $\mu = 0$; its type depends on the value of $\theta(0)$. If $\lambda^n(0) \neq 1$ for $n = 1, 2, 3$, or 4, then an invariant torus emerges as μ passes through zero. If $\lambda^n(0) = 1$ for $n = 1, 2, 3$, or 4 then a special case called resonance occurs. This case has been studied in [11] and [20]; in general the bifurcation is from one periodic solution to another.

The actual analysis of a bifurcation from a periodic solution of a system such as (1.4) must be done numerically since the periodic solution cannot be determined analytically. All numerical integration of the equations in the present study was carried out using the subroutine GEAR [7]. The local error parameter was set at 10^{-7}. This was found to give a solution which converged to an accuracy of 10^{-5} uniformly over the integration interval.

The first step in the analysis is to determine the Poincaré map Ψ. In order to do this a hyperplane Σ in the phase space is selected and the equation is integrated to the next value of t at which the solution crosses that hyperplane in the same

direction. In the present case the hyperplanes selected are those described by the equation $c =$ constant for some constant. In the present problem, with g_I as the bifurcation parameter, the largest amplitude deviation from the periodic solution occurs when $V(t)$ is near its maximum value. For this reason the values of the constant defining the hyperplane are always chosen so that the variable V is near its peak (this also ensures that the hyperplane Σ is transversal to the orbit Γ of the solution). Starting from Σ, the solution is integrated numerically until the first value of t after the solution passes through Σ again moving in the same direction. The value of t such that $c(t) = c(0)$ is then found by Newton iteration with the convergence criterion $|c(t) - c(0)| \leqslant 10^{-5}$. For values of g_I less than the bifurcation value this procedure gives solutions which are periodic in all components to at least four places, except near the bifurcation value, where the periodicity of the V component deteriorates to about two places.

The next step in the analysis is to determine the Jacobian matrix $D\Psi$ at the fixed point q_0. This is accomplished using the midpoint rule. The solution is integrated numerically from initial values $q_0 + he_i$ and $q_0 - he_i$, $i = 1, \ldots, 4$, where the e_i are unit vectors in Σ in the direction of the components V, y_I, x_T, and x_K. In order to normalize the V component the actual differential equation which is integrated involves the variable $\tilde{V} = V/75$ (the value of the potassium equilibrium potential is -75 mV, as given in the Appendix). The midpoint rule for the components of the Jacobian $D\Psi(q_0)$ is

$$\frac{\partial \Psi_i}{\partial q_j} = \frac{\Psi_i(q_0 + he_j) - \Psi_i(q_0 - he_j)}{2h} + 0(h^2).$$

Values of $D\Psi(q_0)$ are obtained for $h = 5 \times 10^{-4}$ and $h = 2.5 \times 10^{-4}$, and Richardson extrapolation is performed on each element to obtain the matrix $D\Psi(q_0)$ used in the analysis. The elements of the extrapolated matrix generally agree to within a relative error of 10^{-2} with those of the matrix computed using the value $h = 2.5 \times 10^{-4}$.

The next step in the analysis is to determine the eigenvalues of the matrix $D\Psi(q_0)$. This is accomplished using the IMSL subroutine EIGRF [10]. Once the eigenvalues are obtained, the bifurcation value of g_I may be determined by searching for that value of g_I such that $|\lambda(g_I)| = 1$. Although this procedure may be automated by using a root finder, for this initial analysis the procedure was carried out by hand (a difficulty with the automated procedure is that the Poincaré map is not computed for values of g_I greater than the bifurcation value). The results obtained numerically for values of g_I near the bifurcation value are shown in Table 1.

The other three eigenvalues each have modulus less than 10^{-3}. Thus our results indicate that a single eigenvalue crosses the unit circle at $\theta = 0$ in the complex plane at approximately $g_I = 1.37052$. The validity of this result was checked by computing the value of the largest eigenvalue at $g_I = 1.37052$ using several values for the constant defining the manifold Σ. The reason for the anomalous value of λ at $g_I = 1.370517$ is not known.

The numerical results summarized in Table 1 indicate that the bifurcation at $g_I = 1.37052$ is a case of resonance, with a single characteristic multiplier crossing

Table 1

g_I	λ
1.370500	0.603
1.370510	0.654
1.370515	0.743
1.370517	0.643
1.370519	0.819
1.370520	1.03

the unit disc at $\lambda = 1$. There are three types of bifurcation which most commonly occur in this case [6]. These are shown schematically in Fig. 5, together with simple example of a Poincaré map displaying this bifurcation. The solid curves represent stable equilibrium solutions (i.e. solutions of $\Psi(q;\lambda) = q$), the dotted curves represent unstable equilibrium solutions, and the dashed curve in Fig. 5a represents a presumed continuation of the solution through another bifurcation point where the solution becomes stable.

A second stable solution is not observed for values of g_I larger than the bifurcation value, as would be required for the bifurcation of Fig. 5c. There is no observed hysteresis, or overlap of stable periodic solutions with different behavior, as would be required for the bifurcation of Fig. 5a. Therefore the conjecture is that the bifurcation is of the type shown in Fig. 5b, an exchange of stability between two periodic solutions of the same period.

Figure 6 is a plot of the solution at a value of g_I just below the bifurcation value. Just above the bifurcation value the integration scheme cannot accurately compute a periodic solution, as shown in Fig. 7. As g_I is increased from the critical value the numerical stability of the solution returns as shown in Fig. 8. From a single spike per "burst" as in Fig. 8, the number of spikes per burst increases with g_I so that there are two spikes per burst at $g_I = 1.75$ (Fig. 9), three per burst at $g_I = 2.0$, and so on. These additional spikes are apparently caused by successive bifurcations similar to the initial one producing the first spike. The exact nature of the initial and subsequent bifurcations is currently under investigation using a related but much simpler system of equations.

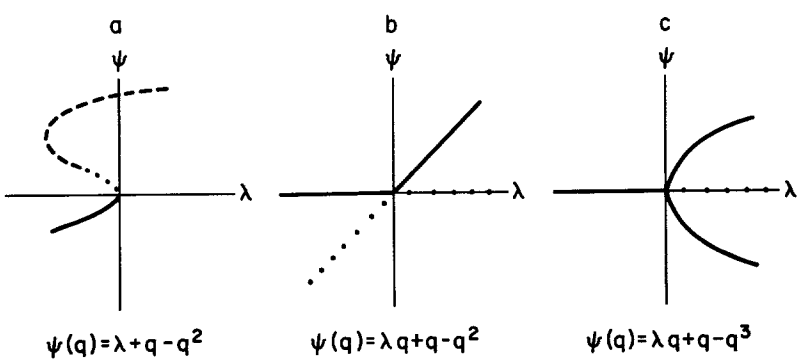


Fig. 5

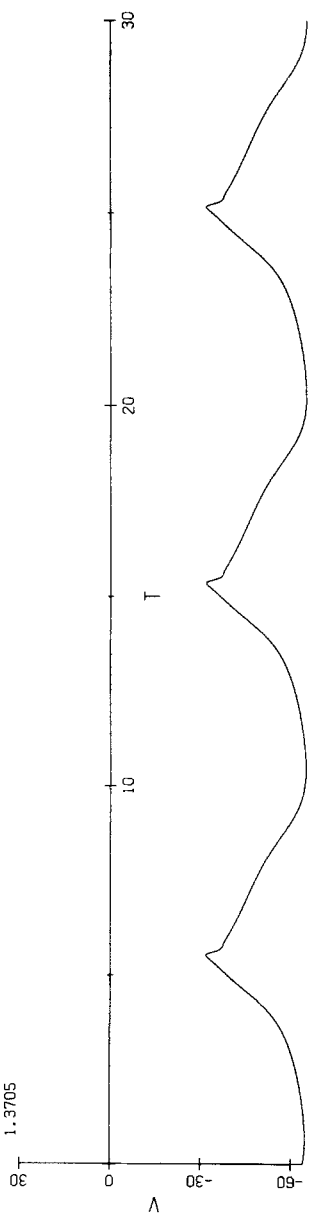


Fig. 6

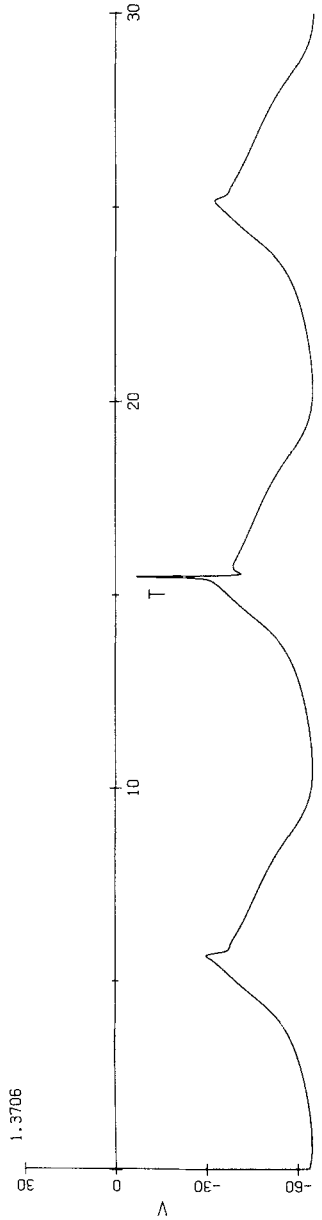


Fig. 7

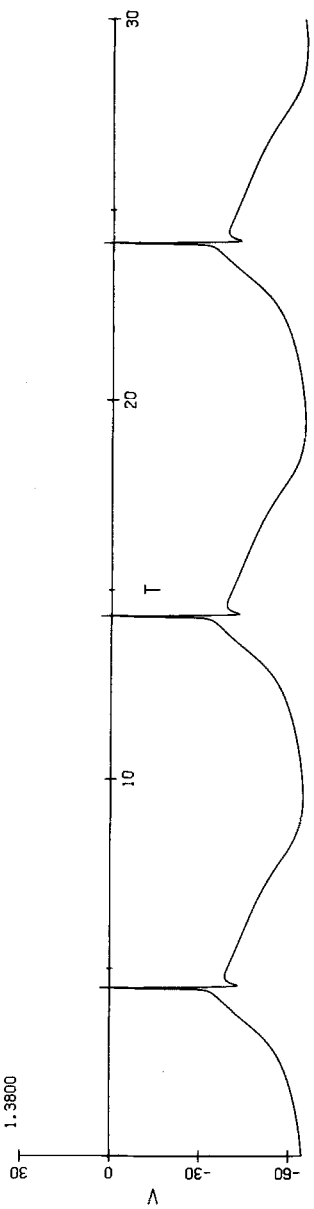


Fig. 8

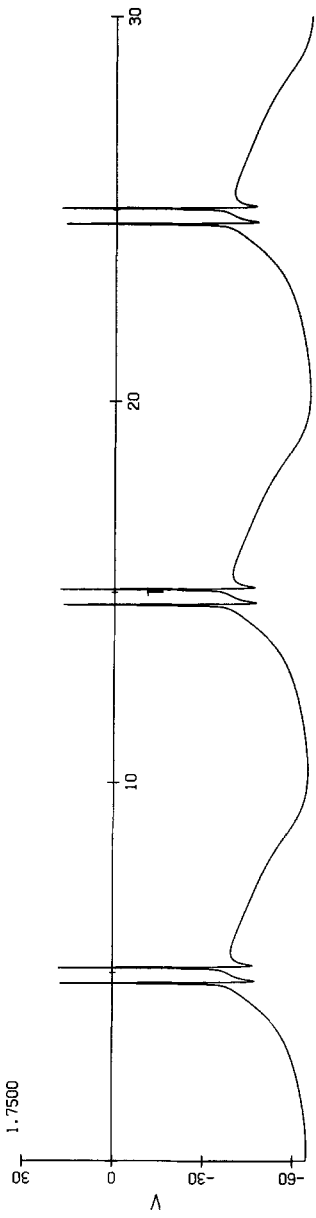


Fig. 9

4. Discussion

In those invertebrate neurons which display spontaneous activity such as bursting, this activity may frequently be altered by the introduction of pharmalogical agents or injected current. One form of such a transition is from the bursting mode to the slow wave mode as schematized by Figs. 1 and 2. In *Aplysia* cells, a third mode called *beating* which consists of a regular train of impulses may also be induced by the application of the metabolic inhibitor ouabain or by the injection of depolarizing current [12]. The transition from one form of solution to another may be observed in the laboratory, and the nature of this transition provides a basis of comparison of models for nerve membrane behavior. The nature of the bifurcation or branching of the solutions of the model is of crucial importance in this comparison.

In the present study the bifurcation parameter for the branching of periodic solutions is the inward spike conductance g_I. As has been mentioned, the conductance modelled by this parameter may be altered experimentally by the application of TTX. It is therefore of interest to compare the behavior of the model near the bifurcation point with that observed in the real system upon application of TTX. Mathieu and Roberge [14, Fig. 3] provide a record of the transition from normal bursting to slow oscillation in cell L6 of the *Aplysia* abdominal ganglion (as has been mentioned, the bursting waveform of the model closely resembles that of this cell). This transition takes approximately ten minutes.

Based on the results of Sections 2 and 3 we would expect that the initial response to TTX will be a reduction in the number of spikes per burst until each "burst" consists of a single spike. This spike should then be reduced in amplitude to reveal the underlying slow wave. Since the bifurcation is a resonance case with a single eigenvalue passing through unity, the period of the oscillation should remain approximately constant during the transition from single spike bursting to slow wave, which is the transition through the branching of solutions. Each of these behaviors is displayed in Fig. 3 of [14]. The number of spikes per burst passes successively from eleven to one, and the last spike before the emergence of the slow wave is reduced in amplitude. The initial slow waves have subthreshold oscillations similar to those of Fig. 6, and these oscillations are gradually reduced in size until a wave similar to that of Fig. 1 remains. This resemblance of the transitionary state between the model and the experiment lends credence to the model.

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Appendix

Here we give the values of the various parameters and functions used in the model. Referring to equations (1.4), the following general assumptions are made.

i) $g_i \geqslant 0, i = I, T, K, P, L$ (note, however, that this assumption was violated in the computation of the bifurcation diagram of Fig. 5).

ii) $V_K < V_L < V_I < V_{Ca}$.

iii) K_P, K_c, ρ, ζ all positive.

Table 2

Parameter	V_I	V_K	V_L	V_{Ca}
Value	30	− 75	− 40	140

Parameter	g_I	g_T	g_K	g_P	g_L
Value	a	b	0.3	0.03	0.003

Parameter	K_P	K_c	ρ	τ_{xT}
Value	0.5	0.0085	0.0003	235

iv) $10 < \tau_{xT}, \tau_{xK}(V), \tau_{yI}(V) \ll \rho^{-1}$ uniformly in V.

v) The functions $s_i(V)$, $i = I, T, K$ (resp. $z_I(V)$) are continuously differentiable over the real line, monotonically increasing resp. decreasing), with a single inflection point, and satisfying

$$\lim_{V \to \infty} s_i(V) = 1 \qquad \left(\text{resp. } \lim_{V \to \infty} z_I(V) = 0\right)$$

and

$$\lim_{V \to -\infty} s_i(V) = 0 \qquad \left(\text{resp. } \lim_{V \to -\infty} z_I(V) = 1\right).$$

The specific values chosen for the numerical study in Section 3 and the end of Section 2 are given in Table 2. The dimensional units are as in [16].

The function $s_T(V)$ has the form $s_T(V) = [\exp(0.15[-50 - V]) + 1]^{-1}$. The functions $s_i(V)$, $i = I, K$, and $z_I(V)$ are rescaled from the Hodgkin-Huxley functions [8]. Let V_{Na}^H and V_K^H be the Hodgkin-Huxley equilibrium potentials, which are 115 mV and − 12 mV respectively. Then let $V_s = aV + b$ where

$$a = (V_{Na}^H - V_K^H)/(V_I - V_K) \qquad \text{and} \qquad b = (V_{Na}^H V_K - V_I V_K^H)/(V_I - V_K).$$

Let

$$\alpha_m(V) = 0.1[50 - V_s]/[\exp\{(50 - V_s)/10\} - 1],$$

$$\beta_n(V) = 4\exp\{(25 - V_s)/18\},$$

$$\alpha_h(V) = 0.07\exp\{(25 - V_s)/20\},$$

$$\beta_h(V) = (\exp\{(55 - V_s)/10\} + 1)^{-1},$$

$$\alpha_n(V) = 0.01[55 - V_s]/[\exp\{(55 - V_s)/10\} - 1],$$

$$\beta_n(V) = 0.125\exp\{(45 - V_s)/80\}.$$

Then the functions $s_i(V)$, $z_I(V)$, and $\tau_i(V)$ are given by

$$s_I(V) = \alpha_m(V)/[\alpha_m(V) + \beta_m(V)],$$

$$s_K(V) = \alpha_n(V)/[\alpha_n(V) + \beta_n(V)],$$

$$z_I(V) = \alpha_h(V)/[\alpha_h(V) + \beta_h(V)],$$

$$\tau_{xK}(V) = 12.5/[\alpha_n(V) + \beta_n(V)],$$

$$\tau_{yI}(V) = 12.5/[\alpha_h(V) + \beta_h(V)].$$

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